

Project Proposal Report

Passive Wave-Driven Hydrofoil Propulsion System

Team WaveX

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1.Project Overview

1.1 Project Title and Introduction

The project explores the use of ocean wave energy as a passive source of propulsive force for marine vessels, with direct relevance to small-scale fishing communities in Sri Lanka.

The concept centres on a submerged hydrofoil that oscillates under wave forcing without any external energy input. The unsteady hydrodynamic forces generated during this oscillation can, under suitable conditions, produce a net forward thrust. This system is intended as an auxiliary propulsion aid, capable of reducing fuel consumption and operating costs while generating zero emissions during operation.

1.2 Purpose and Need

Ocean waves represent a continuous, renewable, and largely untapped source of mechanical energy available at the ocean surface. Despite growing global emphasis on low-emission marine technologies, the vast majority of vessels in the small-scale fishing sector continue to depend exclusively on fossil fuel engines for propulsion. This presents a clear technological gap: a simple, low-cost, and passive system that can harness wave energy to supplement conventional propulsion.

In the Sri Lankan small-scale fishing industry, fuel represents one of the largest operational expenditures. Fishing vessels typically operate in coastal waters where wave activity is persistent, making them ideal candidates for wave-assisted propulsion. Even a modest reduction in fuel consumption estimated at 10 to 15% at fleet scale would yield meaningful economic savings and a measurable reduction in CO₂ emissions.

The need for this project is therefore twofold: to develop a fundamental understanding of how a passive oscillating hydrofoil generates thrust under wave forcing, and to assess the engineering feasibility of such a system through controlled laboratory experimentation.

2. Analysis of Existing System

2.1 Review of Current Propulsion Technologies

Several alternative and supplementary propulsion technologies are currently available or under development for marine vessels. The following table provides a detailed comparison of their specifications, limitations, and how the proposed solution addresses those shortcomings.

Table 01: Comparison of Existing Propulsion Technologies and Proposed Solution

Alternative System	Key Specifications and Limitations	How the Proposed Solution Addresses This
Hybrid / Diesel-Electric Propulsion	High capital cost, requires onboard charging infrastructure, heavy battery systems, complex integration, not suitable for low-income fishing fleets.	The proposed system requires no electrical infrastructure, no battery, and no active drive system. It operates purely through passive mechanical interaction with waves.

Solar-Powered Boats	Heavily dependent on sunlight, limited to daylight operation, reduced efficiency under cloud cover, panels require maintenance, unsuitable for all-weather use.	Wave-driven propulsion is independent of weather or time of day. Waves persist regardless of sunlight, making it more reliable for coastal fishing operations.
Optimized Propellers and Nozzles	Improves fuel efficiency by 5–10% but still relies entirely on fuel-driven engines, provides no renewable energy source, does not reduce fuel dependence fundamentally.	The proposed system supplements or partially replaces engine thrust using wave energy, directly reducing fuel dependence rather than simply improving engine efficiency.
Sail-Assisted Propulsion	Highly weather-dependent; requires skilled operation, complex rigging, impractical for small fishing boats with limited deck space, unsuitable in calm conditions.	The hydrofoil system requires no crew skill to operate, has no exposed above-deck hardware, and functions passively at all times the vessel is at sea in wave conditions.

2.2 Key Insight from Recent Research

The most directly relevant prior work is the experimental study by Wang *et al.*^[2] which investigated a fully passive hydrofoil (spring-loaded in both heave and pitch) in a controlled wave tank. Their findings confirm that a fully passive foil with restoring springs can generate net thrust from wave energy alone, that lower pitch spring stiffness produces greater thrust, and that the load signal exhibits multi-harmonic behavior consistent with energy harvesting. Importantly, Wang *et al.* did not perform flow visualization (observing the reverse von Kármán vortex street experimentally remains an open contribution, which this project directly addresses).

3. Project Objective and Methodology

3.1 Project Objectives

The following five objectives define the scope and deliverables of this project. Each is clearly justified in terms of its contribution to the overall aim.

Table 02: Project Objectives and Justification

Objective	Description and Justification
Objective 1	Design a simple hydrofoil and passive oscillation mechanism based on fundamental hydrodynamic considerations. Justified by the need to establish a theoretically grounded starting point - foil geometry, pivot location, and spring stiffness must be selected using established hydrodynamic principles (Strouhal number, natural frequency, added mass) rather than trial-and-error.

Objective 2	Build an experimental setup capable of producing sustained oscillatory motion under wave forcing. Justified because without a reliable physical apparatus that responds repeatably to wave input, no meaningful force data can be collected. The setup must sustain oscillation near resonance for the duration of each test run.
Objective 3	Measure the forces acting on the system and determine the net thrust produced. Justified as the primary performance metric - net thrust directly quantifies whether the system generates propulsion and how pitch spring stiffness and wave conditions influence it.
Objective 4	Visualize the flow around the oscillating foil and identify key flow features, specifically the reverse von Kármán vortex street. Justified as an experimental investigation to identify vortex shedding dynamics in the hydrofoil wake through time-resolved analysis of observable flow signatures using tracer-assisted visualization, image-based signal extraction, and validation via Strouhal scaling.
Objective 5	Interpret the observations in terms of unsteady fluid dynamics and wave-structure interaction. Justified because raw force and flow data only become meaningful when connected to the underlying physics - Strouhal number, added mass effects, phase relationships between heave and pitch, and wave orbital motion.

3.2 Methodology

The methodology is structured around a quasi-two-dimensional experimental approach. The system is restricted to dominant heave motion with limited passive pitch in order to isolate the essential thrust-generation physics while maintaining fabrication feasibility within a single semester. The following steps describe the approach in sequence.

Step 1 — Hydrodynamic Design & Sizing

- Select foil geometry: NACA 0012 - a well-documented symmetric foil with known lift-drag characteristics, suitable for oscillating foil applications.
- Determine target heave natural frequency using ,

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m_{eff}}}$$

where, m_{eff} = foil mass + added mass

$$m_{added} = C_a \times \rho_{water} \times V_{displaced}$$

C_a (added mass coefficient) ≈ 1 for heave

- Select heave spring stiffness (k) to achieve near-resonance at the target wave frequency.
- Select pitch spring stiffness based on Wang et al. guidance: two values will be tested (low stiffness and higher stiffness) to characterize the stiffness effect on thrust.
- Verify thrust generating conditions using Strouhal number: $St = fA / U$

Step 2 — CAD Design & Fabrication

- Model the foil, heave spring system and carriage rail in SolidWorks.
- Fabricate NACA 0012 foil in fiber. Machine metal components (carriage) in the workshop.
- Assemble the mechanism with compression heave spring (inner/outer rod arrangement) and extension pitch springs at the pivot.
- Include physical travel stops to prevent over-extension and protect the foil during resonant oscillation.

Step 3 — Bench Testing

- Verify natural frequency of the assembled mechanism in air using impulse excitation and accelerometer or manual observation.
- Check spring preloads and pivot freedom before immersion.

Step 4 — Wave Flume Integration & Force Measurement

- Mount the carriage on rails over the wave flume. Fix the carriage in the surge direction for Phase 1 (force measurement).
- Set wave conditions.
- Measure real-time thrust (surge direction) using a load cell connected via LabVIEW data acquisition. Record 30+ seconds of stable signal per test run. Repeat each condition three times.

Step 5 — Flow Visualization

- Release dye or tracer particles upstream of the foil during oscillation.
- Record wake structure using a high-definition camera from the side of the flume.
- Identify presence and structure of the reverse von Kármán vortex street (alternating vortices arranged to induce a downstream momentum jet as the signature of thrust production).
- Compare vortex patterns under different pitch spring stiffness conditions.

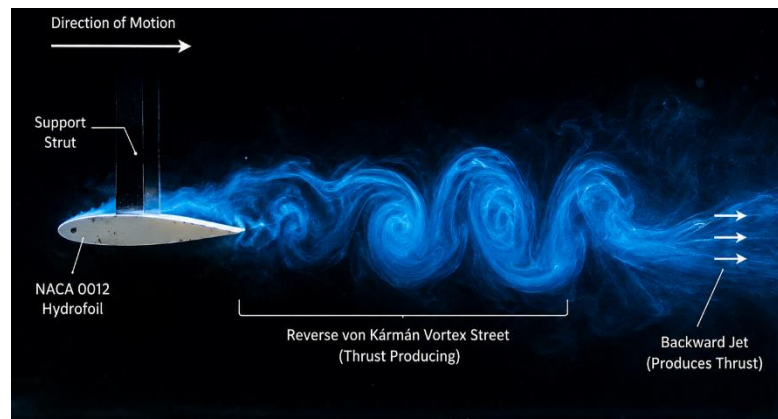


Figure 01: Flow Visualization of Thrust-Producing Reverse von Kármán Vortices

Step 6 — Data Analysis & Interpretation

- Correlate measured thrust with Strouhal number, spring stiffness, and observed vortex structure.
- Interpret heave-pitch phase relationships and their effect on effective angle of attack and thrust generation.
- Compare results against Wang *et al.*^[2] and available semi-passive foil literature.

3.3 Software Tools

Table 03: Software Tools Selected for Design and Analysis

Software	Planned Use
SolidWorks	3D CAD modelling of hydrofoil geometry, suspension rod, spring housing, and carriage. Used to develop design models and simplified dimensioned drawings to communicate specifications for fabrication guidance.
LabVIEW	Real-time data acquisition from the load cell during wave flume testing. Signal logging, filtering, and export for post-processing.

4. Project Planning & Timeline

4.1 Project Milestones

The project is organized into four sequential milestones. Each milestone has a defined scope and clear deliverables to ensure steady progress within the allocated semester timeframe.

Table 4: Project Milestones and Deliverables

Milestone	Scope and Deliverables
Milestone 1 Research & Conceptualization (29 Apr – 13 May 2026)	Complete literature review including Wang <i>et al.</i> ^[2] and related semi-passive foil studies. Finalize project concept, identify wave parameters, select foil geometry (NACA 0012), and determine target spring stiffness range from resonance calculations. Deliverable: Finalized concept design brief.
Milestone 2 Detailed Design & Fabrication (14 May – 10 Jun 2026)	Complete SolidWorks CAD models for all components. Fabricate the foil (fiber), mechanical linkage (machined aluminium), spring assemblies, and carriage. Perform bench testing to verify natural frequency and mechanical function. Deliverable: Fully assembled and bench-tested experimental rig.
Milestone 3 Wave Flume Integration & Experimental Campaign (11 Jun – 15 Jul 2026)	Mount rig in the wave flume. Run Phase 1 (force measurement, fixed carriage) and Phase 2 (flow visualization, free carriage) experiments for two pitch spring stiffness values. Conduct baseline locked-foil tests. Collect minimum 30 s of stable force data per condition, three repetitions each. Deliverable: Complete experimental dataset.
Milestone 4 Data Analysis & Documentation (16 Jul – 7 Aug 2026)	Post-process force data (phase-averaging, spectral analysis). Analyze flow visualization footage for vortex structure. Interpret results in terms of unsteady hydrodynamics. Write final report and prepare project demonstration. Deliverable: Final report and demonstration.

4.2 Project Timeline

The Gantt chart below summaries the planned schedule. The current status indicates that Milestone 1 (Research & Conceptualization) is almost completed. Detailed design begins 14 May 2026, keeping the project on track for final reporting by 7 August 2026.

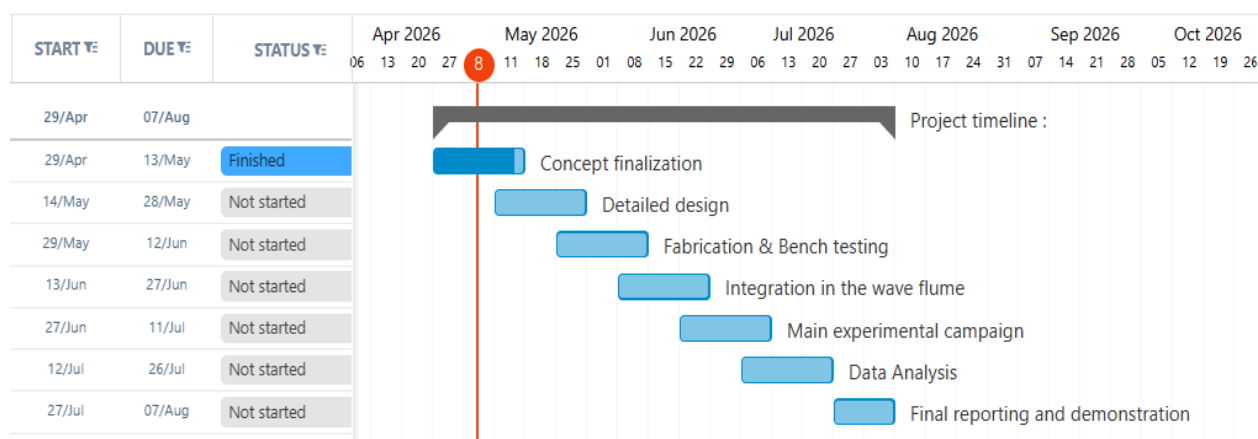


Figure 02: Gantt chart of the planned schedule

5. Impact Analysis

The proposed project has measurable implications across three dimensions: economic, social, and environmental. These are considered in the context of the primary target application, Sri Lanka's small-scale coastal fishing sector.

5.1 Economic Impact

Fuel costs represent one of the most significant operational burdens for small-scale fishing vessels. A wave-assisted propulsion system operating at even 10–15% thrust assistance would translate directly into proportional fuel savings over extended voyages. At fleet scale across Sri Lanka's several thousand registered small fishing vessels, cumulative savings would be economically significant. The proposed system has minimal capital cost. The primary components are a foil, standard springs, and a simple mechanical linkage making it accessible to fishing communities with limited capital.

5.2 Social Impact

Reduced operating costs directly improve the financial resilience of fishing households, which in Sri Lanka's coastal communities often operate at thin margins subject to fuel price volatility. A passive, low-maintenance system that requires no specialist skills to operate is particularly appropriate for this user group. The system requires no modification to the vessel's existing engine or propeller, enabling straightforward retrofitting.

5.3 Environmental Impact

By reducing fuel consumption, the system directly reduces CO₂ and particulate emissions from the fishing fleet. Wave energy is a clean, renewable resource with no fuel cost and no emissions during operation. The system aligns with Sri Lanka's national commitments on reducing maritime emissions and with the global emphasis on low-emission coastal transport. Unlike solar or wind systems, wave energy is available day and night, making it a more consistent supplementary source.

6. References

- (1) Isshiki, H. and Murakami, M. (1983) 'A theory of wave devouring propulsion (3rd report)', *Journal of the Society of Naval Architects of Japan*, 1983(154), pp. 118–128.
- (2) Wang, J., Xing, J., Stagonas, D., Hart, P., Zhang, C., Yang, J. and Yang, L. (2023) 'Experimental study of dynamic response of passive flapping hydrofoil in regular wave', *Physics of Fluids*, 35(7), 077127.